

Multiple Choice (10 marks)

1. In Python 3, which of the following operator in python performs exponential (power) calculation on operands?
 - (a) **
 - (b) //
 - (c) is
 - (d) not in
2. Which of the following has the greatest potential for compromising a user's personal privacy?
 - (a) A group of cookies stored by the user's Web browser
 - (b) The Internet Protocol (IP) address of the user's computer
 - (c) The user's e-mail address
 - (d) The user's public key used for encryption
3. A starvation-free job-scheduling policy guarantees that no job waits indefinitely for service. Which of the following job-scheduling policies is starvation-free?
 - (a) Round-robin
 - (b) Priority queuing
 - (c) Shortest job first
 - (d) Youngest job first
4. Which of the following is the hexadecimal representation of the decimal number 231_{10} ?
 - (a) 17_{16}
 - (b) E_4_{16}
 - (c) E_7_{16}
 - (d) F_4_{16}
5. Let $l = [1,2,3,4]$. What is $\min(l)$ in Python 3?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4

True / False (10 marks)

Answer True or False

6. True or False: The set of Boolean operators {AND, OR} is not complete for representing all Boolean expressions.
7. True or False: In Python 3, the expression $x \gg 1$ with x set to 8 evaluates to 3, indicating a right bitwise shift operation.
8. True or False: The maximum possible degree of the minimal-degree interpolating polynomial $p(x)$ for $n + 1$ distinct points is $n + 1$.
9. True or False: The language $\{ww \mid w \text{ in } (0 + 1)^*\}$ is context-free and can be accepted by a pushdown automaton.
10. True or False: The values of local variables are typically included in a subroutine's activation record frame for a stack-based programming language.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function for computing square roots using the babylonian method.
12. Write a function to find the longest common prefix in the given set of strings.

End of Paper